```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the Dataset

```
df=pd.read_csv('/Sleep_health_and_lifestyle_dataset.csv')
```

Data Understanding

```
df.head()
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales	5.9	4	30	8	Obese	140/90	85	3000	Sleep

```
df.head()
```


	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales	5.9	4	30	8	Obese	140/90	85	3000	Sleep


```
df.dtypes
```

	0
Person ID	int64
Gender	object
Age	int64
Occupation	object
Sleep Duration	float64
Quality of Sleep	int64
Physical Activity Level	int64
Stress Level	int64
BMI Category	object
Blood Pressure	object
Heart Rate	int64
Daily Steps	int64
Sleep Disorder	object

dtype: object

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 374 entries, 0 to 373
Data columns (total 13 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Person ID                374 non-null    int64
 1   Gender                   374 non-null    object
 2   Age                      374 non-null    int64
 3   Occupation               374 non-null    object
 4   Sleep Duration           374 non-null    float64
 5   Quality of Sleep         374 non-null    int64
 6   Physical Activity Level  374 non-null    int64
 7   Stress Level             374 non-null    int64
 8   BMI Category             374 non-null    object
 9   Blood Pressure           374 non-null    object
 10  Heart Rate               374 non-null    int64
 11  Daily Steps              374 non-null    int64
 12  Sleep Disorder           155 non-null    object
dtypes: float64(1), int64(7), object(5)
memory usage: 38.1+ KB
```

```
df.shape
```

```
(374, 13)
```

```
df.describe()
```

	Person ID	Age	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	Heart Rate	Daily Steps
count	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000
mean	187.500000	42.184492	7.132086	7.312834	59.171123	5.385027	70.165775	6816.844920
std	108.108742	8.673133	0.795657	1.196956	20.830804	1.774526	4.135676	1617.915679
min	1.000000	27.000000	5.800000	4.000000	30.000000	3.000000	65.000000	3000.000000
25%	94.250000	35.250000	6.400000	6.000000	45.000000	4.000000	68.000000	5600.000000
50%	187.500000	43.000000	7.200000	7.000000	60.000000	5.000000	70.000000	7000.000000
75%	280.750000	50.000000	7.800000	8.000000	75.000000	7.000000	72.000000	8000.000000

Check Missing Values

```
df.isnull().sum()
```

	0
Person ID	0
Gender	0
Age	0
Occupation	0
Sleep Duration	0
Quality of Sleep	0
Physical Activity Level	0
Stress Level	0
BMI Category	0
Blood Pressure	0
Heart Rate	0
Daily Steps	0
Sleep Disorder	219

dtype: int64

Sleep disorder has missing values so remove them

```
df=df.dropna()
```

so we droped the all null values in sleep discorder colum

```
df.isnull().sum()
```

	0
Person ID	0
Gender	0
Age	0
Occupation	0
Sleep Duration	0
Quality of Sleep	0
Physical Activity Level	0
Stress Level	0
BMI Category	0
Blood Pressure	0
Heart Rate	0
Daily Steps	0
Sleep Disorder	0

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 155 entries, 3 to 373
Data columns (total 13 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Person ID                155 non-null    int64
 1   Gender                   155 non-null    object
```

```
 2   Age                     155 non-null    int64
 3   Occupation              155 non-null    object
 4   Sleep Duration          155 non-null    float64
 5   Quality of Sleep        155 non-null    int64
 6   Physical Activity Level 155 non-null    int64
 7   Stress Level            155 non-null    int64
 8   BMI Category            155 non-null    object
 9   Blood Pressure          155 non-null    object
 10  Heart Rate              155 non-null    int64
 11  Daily Steps             155 non-null    int64
 12  Sleep Disorder          155 non-null    object
dtypes: float64(1), int64(7), object(5)
memory usage: 17.0+ KB
```

Check Duplicate Rows

```
df.duplicated().sum()
```

```
np.int64(0)
```

it shows zero so no dupicated rows

```
df=df.drop_duplicates()
```

hey we have seen blood pressumes coloum has 2 values so we are slpiting it

spliting coloums

```
df.head()
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
5	6	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia
6	7	Male	29	Teacher	6.3	6	40	7	Obese	140/90	82	3500	Insomnia
16	17	Female	29	Nurse	6.5	5	40	7	Normal	132/87	80	4000	Sleep

```
df[['Systolic', 'Diastolic']] = df['Blood Pressure'].str.split('/', expand=True)
```

```
df
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
5	6	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia
6	7	Male	29	Teacher	6.3	6	40	7	Obese	140/90	82	3500	Insomnia
16	17	Female	29	Nurse	6.5	5	40	7	Normal Weight	132/87	80	4000	Sleep Apnea
...	...	...	...	...	...	...	...	...	...	...	...	...	...
369	370	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
370	371	Female	59	Nurse	8.0	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
371	372	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
372	373	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
373	374	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea

155 rows × 15 columns

we have slipt blood pressure into 2 colums

Check Unique Values

```
df['Gender'].unique()
```

```
array(['Male', 'Female'], dtype=object)
```

```
df['Occupation'].unique()
```

```
array(['Sales Representative', 'Software Engineer', 'Teacher', 'Nurse',
       'Doctor', 'Scientist', 'Lawyer', 'Accountant', 'Engineer',
       'Salesperson'], dtype=object)
```

```
df['Sleep Disorder'].unique()
```

```
array(['Sleep Apnea', 'Insomnia'], dtype=object)
```

Check Value **Counts**

```
df['Sleep Disorder'].value_counts()
```

Sleep Disorder	count
Sleep Apnea	78
Insomnia	77

dtype: int64

```
df['BMI Category'].value_counts()
```

BMI Category	count
Overweight	129
Normal	12
Obese	10
Normal Weight	4

dtype: int64

load the cleanded data

```
df.to_csv("clean_sleep_dataset.csv", index=False)
```

NOW WE WILL DO Exploratory Data **Analysis**

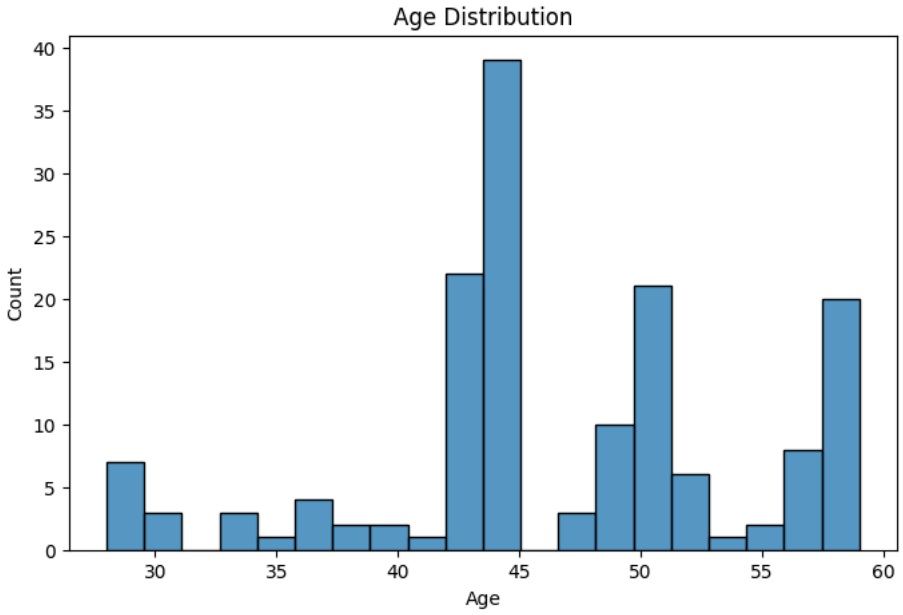
```
df = pd.read_csv("/content/clean_sleep_dataset.csv")
df.head()
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder	Sys
0	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea	
1	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea	
2	6	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia	
3	7	Male	29	Teacher	6.3	6	40	7	Obese	140/90	82	3500	Insomnia	
4	17	Female	29	Nurse	6.5	5	40	7	Normal Weight	132/87	80	4000	Sleep Apnea	

Understand Distribution of **Age**

```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'],bins=20)
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')
```

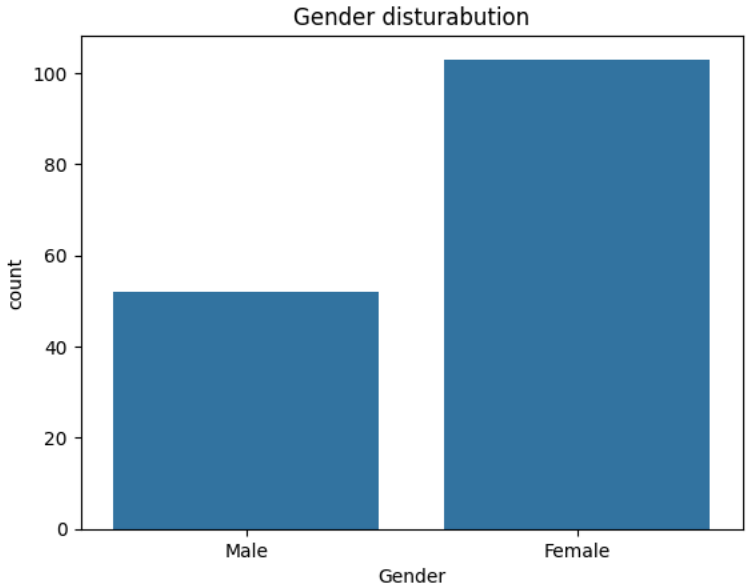
```
Text(0, 0.5, 'Count')
```



observations:The age distribution shows that most individuals fall between 42 and 46 years old. The dataset ranges from approximately 28 to 59 years, with a slight right skew indicating the presence of older individuals beyond the main age group.

```
sns.countplot(x='Gender',data=df)
plt.title('Gender disturabution')
```

```
Text(0.5, 1.0, 'Gender disturabution')
```

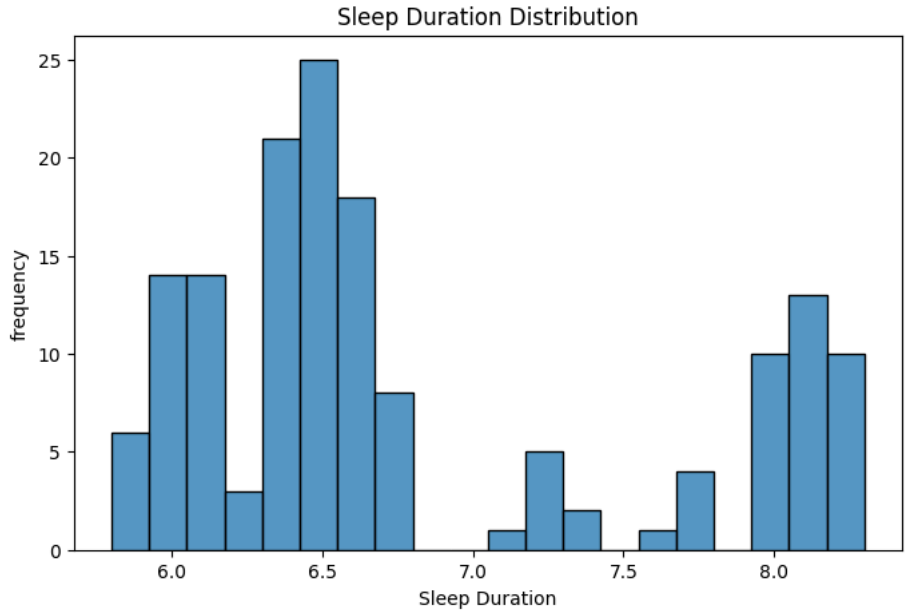


insghts: more females are in the data set ,which mean means have higher stress then compare to men

Sleep Duration Distribution

```
plt.figure(figsize=(8,5))
sns.histplot(df['Sleep Duration'],bins=20)
plt.title('Sleep Duration Distribution')
plt.xlabel('Sleep Duration')
plt.ylabel('frequency')
```

```
Text(0, 0.5, 'frequency')
```



Double-click (or enter) to edit

```
df['Sleep Duration'].mean()
```

```
np.float64(6.812258064516129)
```

```
df['Sleep Duration'].max()
```

```
8.3
```

```
df['Sleep Duration'].min()
```

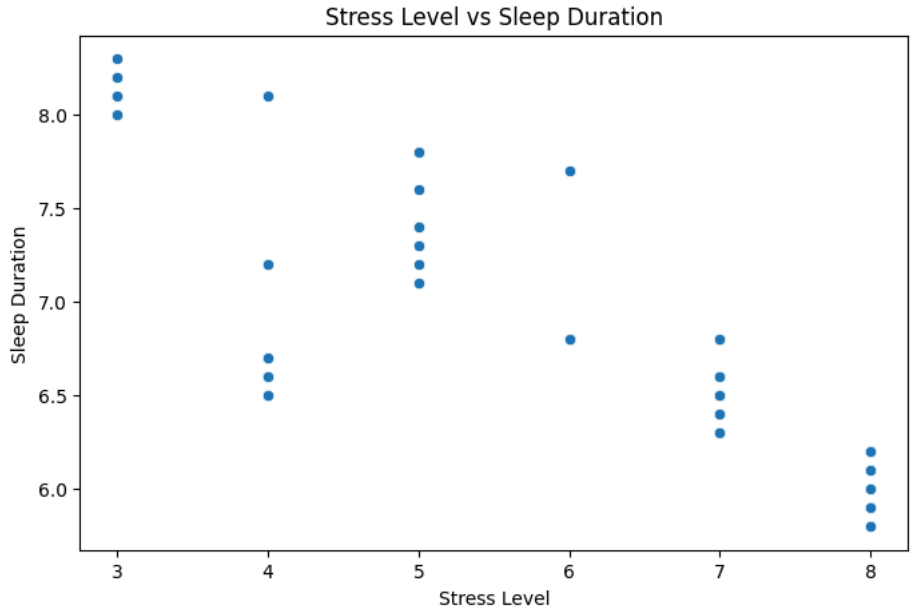
```
5.8
```

insight:L:The sleep duration distribution shows that most individuals sleep between 6 and 6.8 hours, with an average sleep duration of approximately 6.5 hours. The majority of participants sleep less than the recommended 7 hours per night, while a smaller group sleeps around 8 hours.

Stress Level vs Sleep Duration

```
plt.figure(figsize=(8,5))
sns.scatterplot(x='Stress Level',y='Sleep Duration',data=df)
plt.title('Stress Level vs Sleep Duration')
plt.xlabel('Stress Level')
plt.ylabel('Sleep Duration')
```

```
Text(0, 0.5, 'Sleep Duration')
```



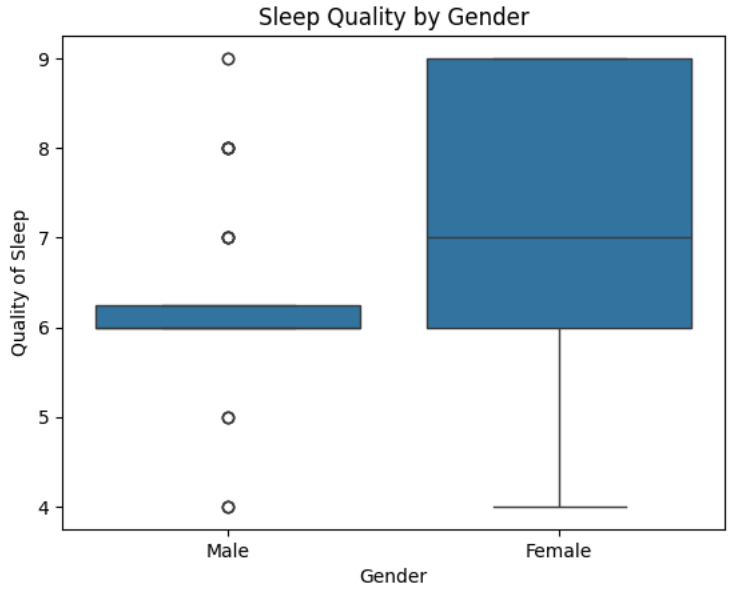
insight:Higher stress levels tend to be associated with shorter sleep duration.

Sleep Quality by Gender

```
sns.boxplot(x='Gender', y='Quality of Sleep', data=df)

plt.title("Sleep Quality by Gender")

plt.show()
```



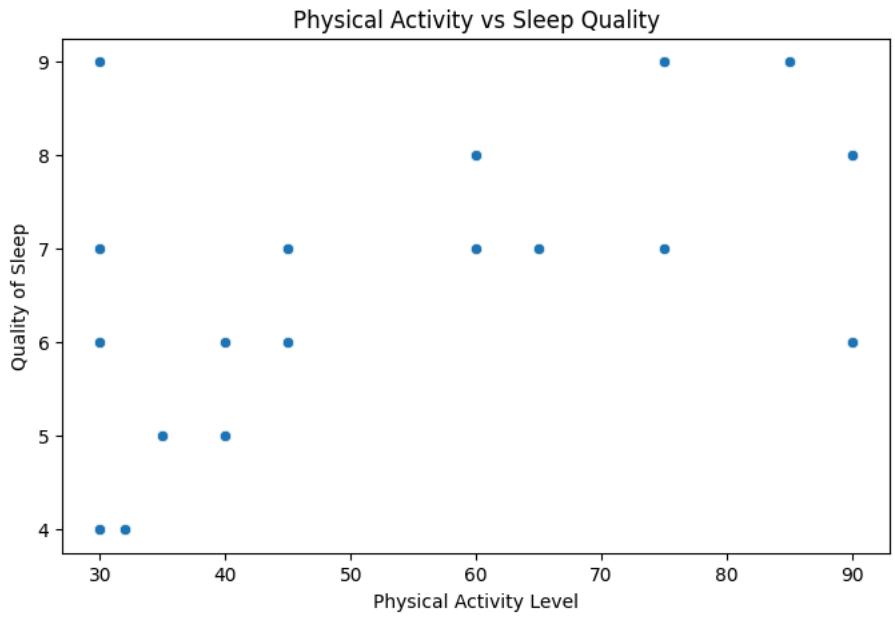
insights:Female participants generally report higher sleep quality, with a median around 7, compared to males whose median sleep quality is around 6. Female sleep quality also shows a wider variation."

```
plt.figure(figsize=(8,5))

sns.scatterplot(x='Physical Activity Level', y='Quality of Sleep', data=df)

plt.title("Physical Activity vs Sleep Quality")
```

```
plt.show()
```



"Higher levels of physical activity appear to be associated with better sleep quality. Individuals with activity levels above 70 generally report sleep quality scores between 7 and 9.

Correlation Heatmap

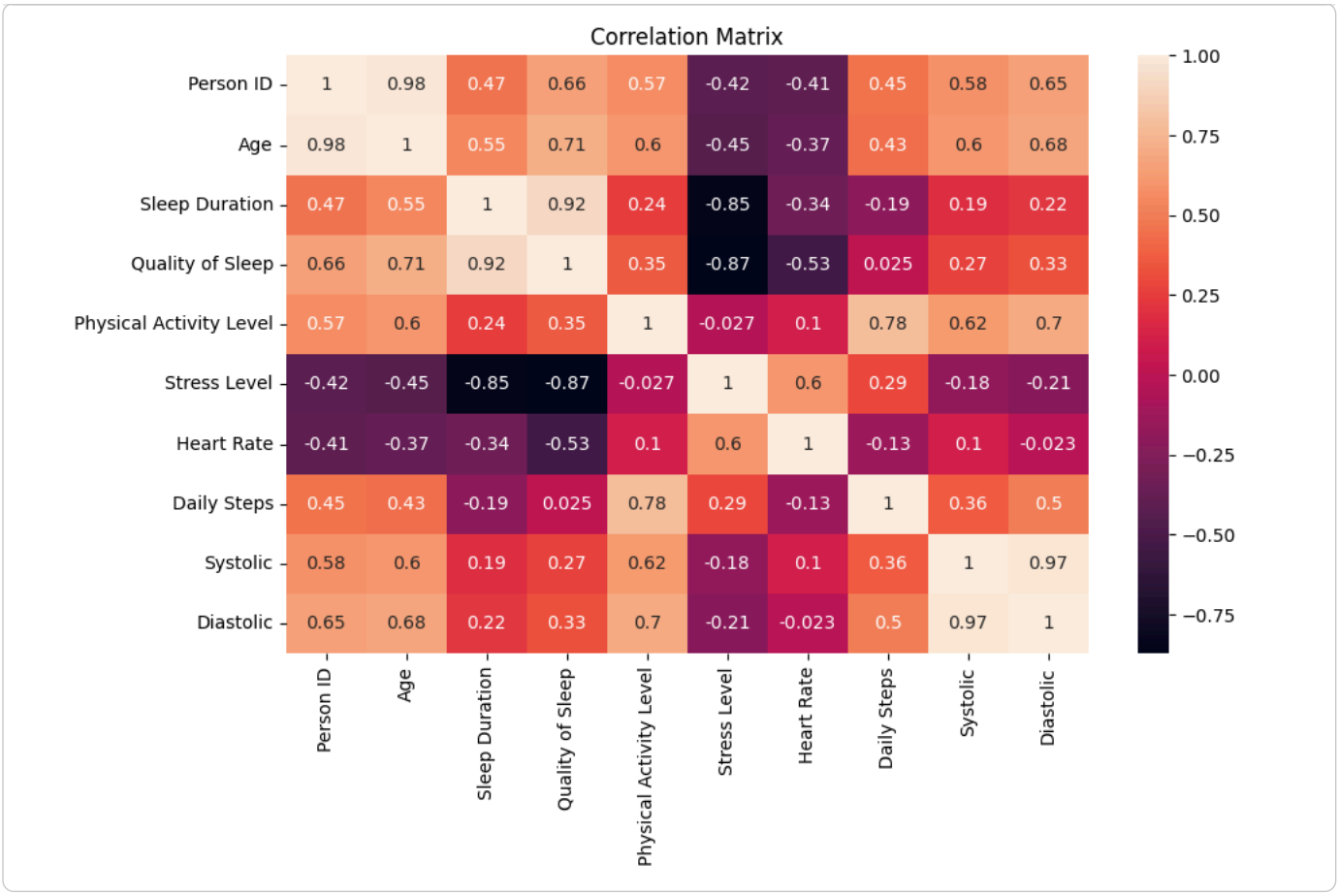
```
corr = df.corr(numeric_only=True)

plt.figure(figsize=(10,6))

sns.heatmap(corr, annot=True)

plt.title("Correlation Matrix")

plt.show()
```



Sleep duration strongly improves sleep quality.

Stress significantly reduces both sleep quality and sleep duration.

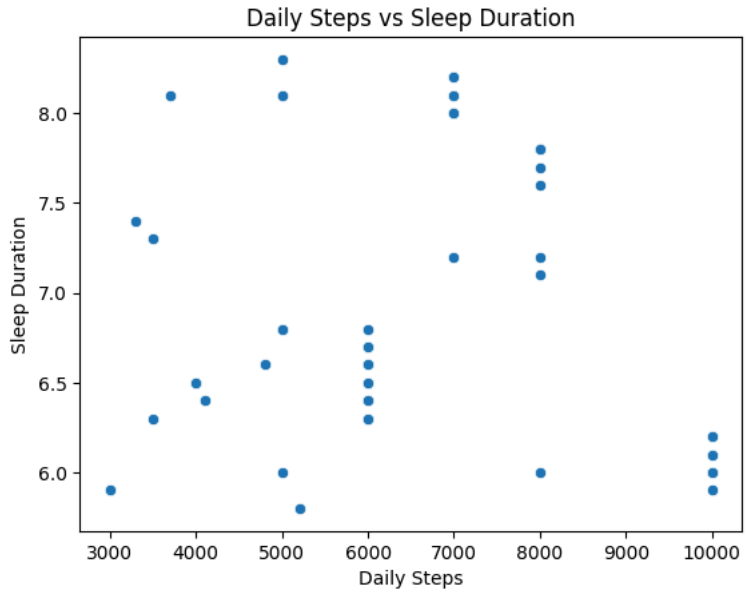
Physical activity is strongly linked with daily steps and better health indicators.

Daily Steps vs Sleep Duration

```
sns.scatterplot(x='Daily Steps', y='Sleep Duration', data=df)

plt.title("Daily Steps vs Sleep Duration")

plt.show()
```



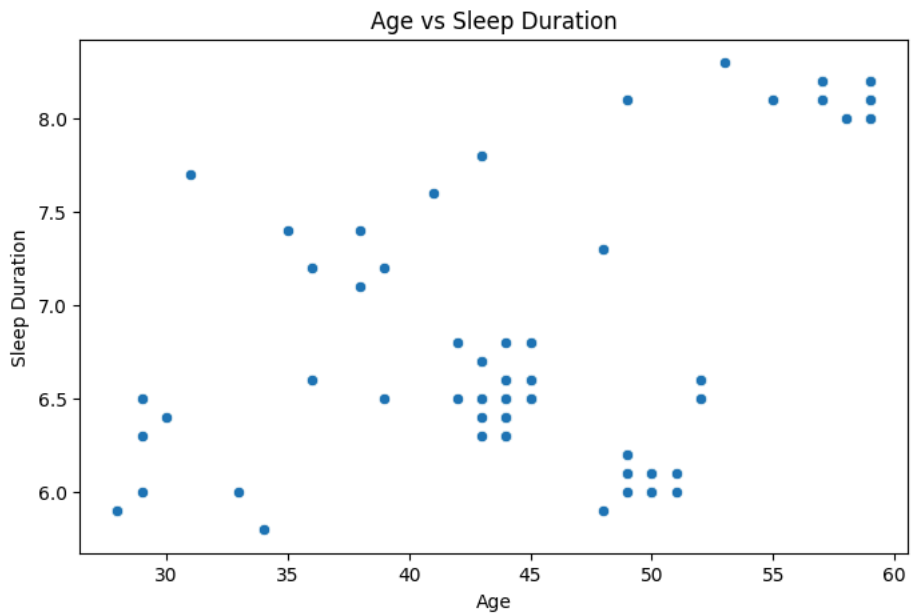
Does Age Affect Sleep Patterns?

```
plt.figure(figsize=(8,5))

sns.scatterplot(x='Age', y='Sleep Duration', data=df)

plt.title("Age vs Sleep Duration")
plt.xlabel("Age")
plt.ylabel("Sleep Duration")

plt.show()
```



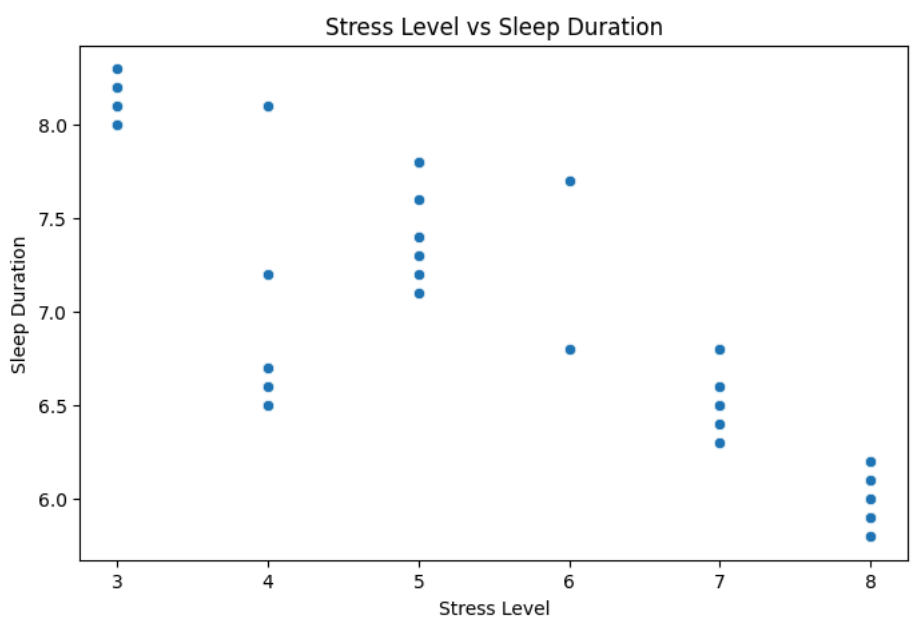
Does Stress Level Affect Sleep Duration?

```
plt.figure(figsize=(8,5))

sns.scatterplot(x='Stress Level', y='Sleep Duration', data=df)

plt.title("Stress Level vs Sleep Duration")

plt.show()
```



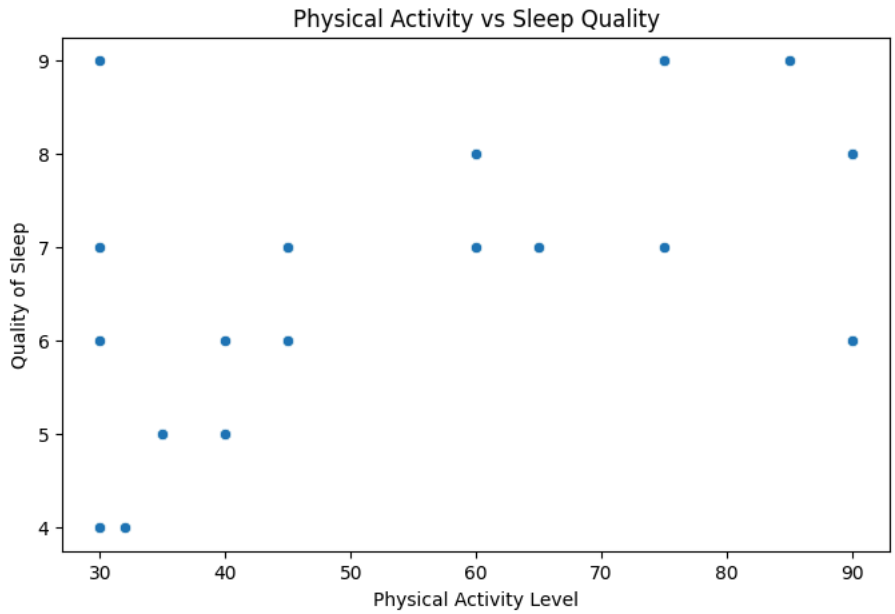
Does Physical Activity Improve Sleep Quality?

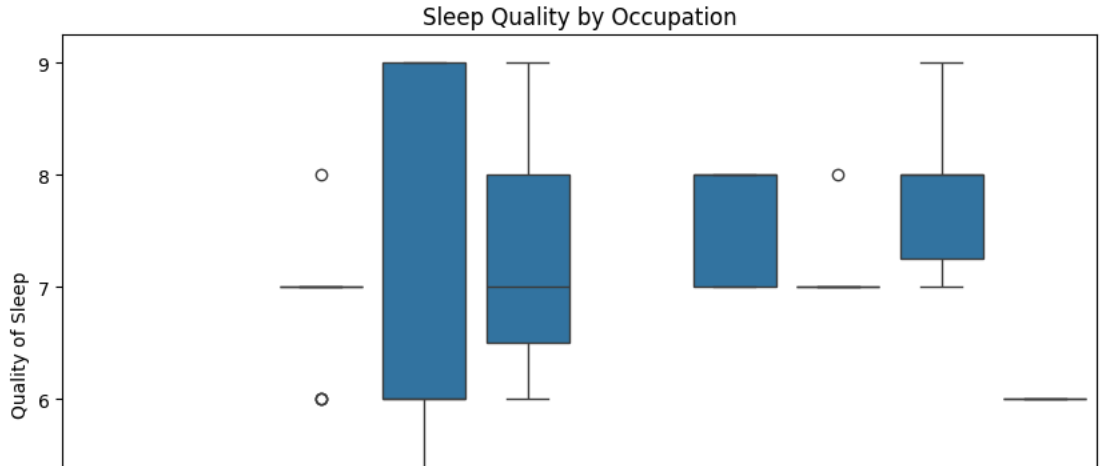
```
plt.figure(figsize=(8,5))

sns.scatterplot(x='Physical Activity Level', y='Quality of Sleep', data=df)

plt.title("Physical Activity vs Sleep Quality")

plt.show()
```



Which Occupations Have Poor Sleep Quality?

```
plt.figure(figsize=(10,6))

sns.boxplot(x='Occupation', y='Quality of Sleep', data=df)

plt.xticks(rotation=45)

plt.title("Sleep Quality by Occupation")

plt.show()
```